

## IN THE PIPELINE

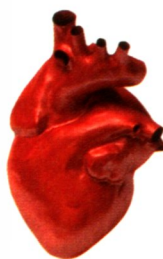
**Water Jet:** High pressure stream of water, used as a cutting tool in industry, is ready to be used for cutting tissues

**Telesurgery:** Surgeons have started using remote-controlled robots to operate on patients across continents

**Go Virtual:** Soon research will enable surgeons to scan patients, create 3-D images and practice before the real surgery

**Robot Brain:** Let a robot use its brains to operate on a virtual patient and use it as a blueprint

## NEW BREAKTHROUGHS IN SURGERY FROM ACROSS THE COUNTRY



**WHEN:** 2008  
**WHAT:** Implanting dual pacemakers for the first time in India.  
**WHERE:** Jaslok Hospital and Research Centre, Mumbai



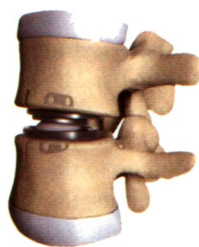
**WHEN:** 2008  
**WHAT:** Robotic chest surgery on thymus gland  
**WHERE:** All India Institute of Medical Sciences, Delhi



**WHEN:** 2008  
**WHAT:** Asia's first artificial heart implant  
**WHERE:** The Narayana Hrudayalaya hospital in Bangalore



**WHEN:** 2008  
**WHAT:** Tissue glue used to fix up intraocular lens  
**WHERE:** The Aggarwal Hospital, Chennai



**WHEN:** 2007  
**WHAT:** Kneecap replacement; artificial discs in spine  
**WHERE:** AIIMS, Bombay Hospital, Hosmat Institute, Bangalore



**WHEN:** 2006  
**WHAT:** Asia's first "Brain Suite" to treat brain tumours  
**WHERE:** Institute of Neuroscience, Max Hospital, Delhi

ments in tiny spaces—without trembling like a tired surgeon's might—and gives him 3D vision."

"None of this would be possible without a vital adjunct—molecular imaging technology," says surgeon and cancer specialist Dr Harit Chaturvedi of the Rajiv Gandhi Cancer Institute and Research in Delhi. "Later versions of Computed Tomography, Magnetic Resonance Imaging and Positron Emission Tomography scanners have ensured that there's no nook or cranny in the human body where a tumour or a disease can hide." Track the change in neurosurgery. "In the past, the difficulty in distinguishing between diseased and healthy brain tissue needed a follow-up MRI a day or so following surgery," says Kale. That delay will be eliminated soon at AIIMS with high field-strength intraoperative MRI—literally bringing imaging to the patient in the surgical suite. On the horizon are the newer, even less-invasive approaches of surgery through natural orifices (Natural Orifice Transluminal Endoscopic Surgery). "The mouth, anus or vagina are used to pass fibre-optic instruments inside the body," says Rao. "The fall-out is minimum scar, pain and trauma."

At the root of all this lies modern India's interplay with the West. State-of-the-art technology was often brought in by Indian doctors who learnt their craft in the West. On some fronts, such doctors far outstripped the West. Most American cardiac surgeons still hesitate to perform "beating heart" surgery, which does not require stopping the heart or using a heart-lung machine. In India, it is increasingly commonplace. Globalisation added the rest of the fillip. Sandeep Sinha, senior healthcare analyst with

market intelligence, Frost & Sullivans (F&S), holds that the entry of cash-rich corporates into the healthcare fray changed the scenario: "They have the wherewithal to mobilise huge resources globally." Most new technology is being adopted fast by them. Being on the cutting edge makes a sensible business proposition.

New technology has also meant spiralling costs, says a F&S report, *Cost of Healthcare in India, 2007*. Consider: if the cost of a standard cataract operation in 2004 was Rs 35,000, with newer lenses coming into the market, it went up to Rs 42,110 in 2006—a 20 per cent hike in two years. If simple X-rays cost Rs 50 in the '60s, ultrasound images in late-'70s cost around Rs 300, CT scans in mid-'80s Rs 2,500, MRI by mid-'90s to Rs 5,000 and now the PET scans cost Rs 25,000. But despite the price war, doctors hold, it matters when someone can walk back home after a gall bladder operation the same day instead of spending a week.

After battling a long-drawn surgery, Kumar enjoys his personal ritual: reloading the hours spent in the OR in his mind over a cup of steaming coffee. Today his mind wanders to the last nine months—of lobbying for a robot, spending a lakh out of pocket for getting trained abroad, not to mention tackling the logistical nightmare of a public institution. They say, new technology will end the era of the 'great man' in surgery. "If only," he smiles to himself. The robot has proved once again that the machine is the slave and the surgeon the master. "Technology will move ahead so long we stick to our mandate of staying at the cutting edge."